

Optimizing your website for generative AI features on Google Search

<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide>

User preferences are rapidly evolving and people are increasingly gravitating to generative AI experiences to help them find information. As we upgrade Search to meet these changing expectations,

this transformation offers new opportunities to reach people who may be more inclined to engage with your site, spend more time with your content, or even convert by becoming a subscriber or making a purchase. This guide is for website owners looking for official best practices from Google Search on how to succeed in generative AI features in Google Search (such as AI Overviews and AI Mode).

Is SEO still relevant for generative AI search?

In short, yes! The [best practices for SEO](<https://developers.google.com/search/docs/essentials>) continue to be

relevant because our generative AI features on Google Search are rooted in our core Search ranking and quality systems. These features rely on AI techniques to highlight content from our Search index, such as:

- **Retrieval-augmented generation (RAG)**: A technique (also known as grounding) used to improve the quality, accuracy, and freshness of AI responses by relying on our core Search ranking systems to retrieve relevant, up-to-date web pages from our Search index. Our systems then review the specific information from those retrieved pages to generate a more reliable and helpful response, showing prominent, clickable links to relevant web pages that support the information in the response.

- **Query fan-out**: A set of concurrent, related queries generated by the model to request more information and fetch additional relevant search results to address the user's query. For example, if the original user's query is "how to fix a lawn that's full of weeds", fanout queries might include "best herbicides for lawns", "remove weeds without chemicals", and "how to prevent weeds in lawn".

> [!IMPORTANT]

> **What about "AEO" and "GEO"?** "AEO" stands for "answer engine optimization" and "GEO" for "generative engine optimization". These are both terms you may see used to describe work specifically focused on improving visibility in AI search experiences. From Google Search's perspective, optimizing for generative AI search is optimizing for the search experience, and thus still SEO. If you're considering third-party "AEO" or "GEO" advice or services, review our [guidance on evaluating third-party SEO advice](<https://developers.google.com/search/docs/fundamentals/third-party-seo>).

Apply foundational SEO best practices to generative AI search

This section focuses on reframing SEO best practices to understand what matters most to AI systems

today and how you can implement them in the context of generative AI search, and ultimately with the goal of improving your website's visibility in both generative AI search experiences and Google Search overall.

Create valuable, non-commodity content for your audience

Creating content that people find unique, compelling, and useful will likely influence your website's presence in generative AI search in the long run more than any of the other suggestions in this guide. While "unique, valuable, good content" can mean different things to different people, content like this generally shares some common attributes, such as:

- **Providing a unique point of view**: Our AI systems take a look at a variety of sources, so it can be helpful to have a unique viewpoint that stands out. For example, a first-hand review provides a unique perspective based on personal experience, whereas a summary of existing content simply restates information already available elsewhere. Create the content yourself based on what you know about the topic, and consider what in-depth experience you can bring to your content. Don't just recycle what others on the internet have already said, or could easily be produced by a generative AI model.
- **Creating non-commodity content that's** [helpful, reliable, and people-first](<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>): Be sure that you're writing non-commodity content that your readers will find helpful and reliable. Commodity content (for example, something like "7 Tips for First-Time Homebuyers") is often based on common knowledge, which could originate from anyone, and typically adds little unique insight for readers. In contrast, non-commodity content (such as "Why We Waived the Inspection & Saved Money: A Look Inside the Sewer Line") provides unique expert or experienced takes that go beyond common knowledge and the ordinary.
- **Organizing content in a way that helps your readers**: Write content for your human audience and make sure the content is well written and easy to follow. People generally appreciate it when web pages are organized by paragraphs and sections, along with headings that provide a clear structure to navigate content.
- **Add high-quality images and video**: Many people appreciate finding images and videos as they search for things online. As with Google Search overall, our generative AI search features can bring in relevant images and video, which means more opportunities for your website to appear beyond web page links. When it makes sense, look for ways to support your textual content with high-quality, relevant images and videos on your pages. If you're already following our [image SEO best practices](<https://developers.google.com/search/docs/appearance/google-images>) and [video SEO documentation](<https://developers.google.com/search/docs/appearance/video>), you're already optimizing for generative AI search.
- **Focus on what your users want, and avoid overdoing it**: While it might be tempting to create separate content for every possible variation of how people might search (for example, by focusing on other queries that people have asked, or fan-out queries), doing so primarily to manipulate rankings or generative AI responses in Google Search violates Google's [scaled content abuse spam policy](<https://developers.google.com/search/docs/essentials/spam-policies#scaled-content>). This is also an ineffective long-term strategy, as a high quantity of pages doesn't make a website higher quality or more relevant to users. Google's AI systems have advanced even further and improved upon our ability to [understand the relevance of pages](<https://blog.google/products-and-platforms/products/search/search-language-understanding-bert/>), even when there is no exact match between the query and the page's primary content.
- **If you're using generative AI tools to assist in content creation**, be sure that your work meets the standards of the [Search Essentials](<https://developers.google.com/search/docs/essentials>) and our [spam policies](<https://developers.google.com/search/docs/essentials/spam-policies#scaled-content>). For more details on our approach, see our [guidance on AI-generated content](<https://developers.google.com/search/docs/fundamentals/using-gen-ai-content>).

You can simplify your approach by focusing on one core principle: focus on what your visitors

would enjoy, find helpful, and feel satisfied with after visiting your website. If you're ever unsure about a decision for your site, ask yourself: "Is this content that my visitors would find satisfying?" If the answer is yes, then you're on the right track, as our systems are designed to connect people with exactly that kind of useful information. For more, check out our guide to [creating helpful, reliable, people-first content](<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>).

Build and maintain a clear technical structure

The way Google Search finds and processes your pages remains the core of how our AI systems access your data. Technical clarity ensures your content is ready for discovery and indexing, and all existing technical [SEO best practices](<https://developers.google.com/search/docs/fundamentals/seo-starter-guide>) continue to be worthwhile, for example:

- ****Meet the Search technical requirements:**** To be eligible to be shown in generative AI features on Google Search, a page must be indexed and eligible to be shown in Google Search with a snippet, fulfilling the [Search technical requirements](<https://developers.google.com/search/docs/essentials/technical>).

In addition to the technical requirements for Search, a site must be [included in Search generative AI features in Search Console](<https://support.google.com/webmasters/answer/16908024>) to be eligible for display in generative AI features on Google Search.

> [!NOTE]

> Just because a page meets all requirements, best practices, and complies with the policies, doesn't mean that Google will crawl, index, or serve its content. Indexing and serving aren't guaranteed. Learn more about [How Search Works](<https://developers.google.com/search/docs/fundamentals/how-search-works>).

- ****Follow crawling best practices**** . To maximize your site's visibility in generative AI search features, ensure your content is crawlable, as Google Search generative AI models use publicly accessible, crawlable content to learn patterns and provide relevant, grounded responses. For very large and frequently updated sites, review our guide to [optimizing your crawl budget](<https://developers.google.com/crawling/docs/crawl-budget>).

- ****When it comes to semantic HTML, focus on human readability and don't worry about perfect code:**** While it's not required to have perfectly semantic HTML (the web in general is not valid HTML, and Google can understand it), it's generally a good idea to try to use semantic HTML when possible, as it helps other types of users, such as screen readers, [parse and navigate your web page more easily](<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide#agentic-experiences>).

- ****If you're using JavaScript, be sure to follow JavaScript SEO best practices**** . Google is able to process content within JavaScript as long as it isn't blocked. That said, working on SEO with a website that uses JavaScript frameworks is generally more complex than when working with other kinds of websites. Make sure to follow the usual [SEO best practices for JavaScript](<https://developers.google.com/search/docs/crawling-indexing/javascript/javascript-seo-basics>).

- ****Provide a [good page experience](<https://developers.google.com/search/docs/appearance/page-experience>)**** for those who arrive at your site. This includes ensuring your site displays well across all devices, reducing latency, and making it easy for people to distinguish your main content from other elements on the page.
- ****Reduce duplicate content:**** Having duplicate content can be a bad user experience and search engines might waste crawling resources on URLs that you don't even care about. If you have time, [try to reduce it](<https://developers.google.com/search/docs/fundamentals/seo-starter-guide#reduce-duplicate-content>).

To discover and diagnose potential technical issues quickly, [verify your site in Search Console](<https://support.google.com/webmasters/answer/9008080>).

For more, check out our [technical guide to

SEO](<https://developers.google.com/search/docs/fundamentals/get-started-developers>)

and [maintaining your website's SEO](<https://developers.google.com/search/docs/fundamentals/get-started>).

Optimize your local business and ecommerce details

Where appropriate, generative AI responses can include product listings, product information, and information about local businesses. Using products like [Merchant

Center](<https://merchants.google.com/>)

(such as [Merchant Center feeds](<https://support.google.com/merchants/answer/11586438>))

and [Google Business Profiles](<https://business.google.com/>) can

help your products and services to be visible in both AI responses and other Google Search results. Learn more about how to

[add and manage your business details on Google

Search](<https://developers.google.com/search/docs/appearance/establish-business-details>).

> [!NOTE]

> Depending on your business type and goals, consider other merchant experiences like [Business Agent](<https://support.google.com/brandprofile/answer/16410382>), which is a conversational experience on Google Search that helps customers chat with your brand.

Mythbusting generative AI search: what you don't need to do

As generative AI search evolves, so have the theories and practices---and sometimes, the misconceptions---surrounding it. While terms like Answer Engine Optimization (AEO) or Generative Engine Optimization (GEO) are common online, many suggested "hacks" aren't effective or supported by how Google Search actually works.

To help you focus on what matters for your website's visibility, we've collected some of the most prominent topics circulating the internet around generative AI and Google Search. Here are a few things you can ignore for Google Search:

- ****LLMS.txt files and other "special" markup**** : You don't need to create new machine readable files, AI text files, markup, or Markdown to appear in Google Search (including its generative AI capabilities), as Google Search itself doesn't use them. Note that Google may discover, crawl, and

index [many kinds of files](<https://developers.google.com/search/docs/crawling-indexing/indexable-file-types>) in addition to HTML on a website: this doesn't mean that the file is treated in a special way.

> [!NOTE]

> It's completely fine if you decide to create and maintain LLMS.txt files (or other similar files) for other services or systems that use these files. Doing so will neither harm nor help your site's visibility or rankings in Google Search, as Google Search ignores them.

- **"Chunking" content:** There's no requirement to break your content into tiny pieces for AI to better understand it. Google systems are able to understand the nuance of multiple topics on a page and show the relevant piece to users. However, sometimes shorter (or longer!) pages can work well depending on your audience and subject matter. There's no ideal page length, and in the end, make pages for your audience, not just for generative AI search.

- **Rewriting content just for AI systems:** You don't need to write in a specific way just for generative AI search. AI systems can understand synonyms and general meanings of what someone is seeking, in order to connect them with content that might not use the same precise words. This means you don't have to worry that you don't have enough "long-tail" keywords or haven't captured every variation of how someone might seek content like yours.

- **Seeking inauthentic "mentions":** Just like the rest of Google Search, our generative AI features can show what's being said about products and services across the web, including in blogs, videos, and forum discussions. However, seeking inauthentic "mentions" across the web isn't as helpful as it might seem. Our core ranking systems focus on [high-quality content](<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>) while other systems [block spam](<https://developers.google.com/search/docs/essentials/spam-policies>); our generative AI features depend on both.

- **Overfocusing on structured data:** Structured data isn't required for generative AI search, and there's no special schema.org markup you need to add. However, it's a good idea to continue using it as part of your overall SEO strategy, as it helps with being eligible for rich results on Google Search.

Measure visibility in generative AI features with Search Console

To measure how your content is performing in generative AI features on Google Search and Discover, use the [Generative AI performance report](<https://support.google.com/webmasters/answer/16984139>) in Search Console. This can help you get an idea of how people are discovering your content through generative AI features on Google Search.

> [!CAUTION]

> Be wary of third-party tools that promise ranking success or claim to use "internal" Google metrics. No third-party tool has access to our internal ranking or AI systems. Use these tools if they help your workflow, but be sure to [evaluate their advice against our official guidance](<https://developers.google.com/search/docs/fundamentals/third-party-seo>).

Explore agentic experiences

AI agents are autonomous systems that can perform tasks on behalf of people, such as booking a reservation or comparing product specifications. These agents can take many forms; for example,

browser agents may access your website to gather the data they need to complete these tasks, such as analyzing visual renderings (like screenshots), inspecting the DOM structure, and interpreting the accessibility tree.

If this is something that's relevant to your business and you have extra time, check out the available agentic experiences and review the guide to [agent-friendly website best practices](<https://web.dev/articles/ai-agent-site-ux>), which gives some insights into how a website can generally prepare for current browser agents. Protocols like [Universal Commerce Protocol](<https://ucp.dev/latest/>) (UCP) are emerging that will allow Search agents to do more.

Next steps: what to focus on

As you continue working on your website, remember that plenty of content thrives in Google Search (including generative AI experiences) without any overt SEO at all, and you don't need to accomplish everything in this guide in order to succeed on Google Search. To recap, here are the key takeaways from this guide:

- ****Apply SEO best practices to generative AI search:**** Continue prioritizing foundational SEO best practices, such as [building a clear technical structure](<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide#build-technical-structure>) and [creating unique, valuable content](<https://developers.google.com/search/docs/fundamentals/ai-optimization-guide#create-valuable-content>); these are the foundation for visibility in generative AI search experiences (and Google Search overall).
- ****Create non-commodity content that's [helpful, reliable, and people-first](<https://developers.google.com/search/docs/fundamentals/creating-helpful-content>):**** Focus on developing unique, expert-led content that provides value beyond common knowledge.
- ****Prioritize effective SEO strategies over "AEO/GEO hacks":**** For Google Search, you can ignore tactics like "chunking" content, creating unnecessary AI text files (like llms.txt), or pursuing inauthentic mentions.
- ****Monitor visibility in Search Console:**** Use the [Generative AI performance report](<https://support.google.com/webmasters/answer/16984139>) to see how your content is performing in generative AI features on Google Search.
- ****Explore agentic experiences****: Stay informed about emerging technologies that allow AI agents to interact with your site, such as browser agents and new protocols.

Stay informed and ask questions

If you want to learn more about SEO, here are some resources that can help you stay on top of changes and new resources we publish:

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| | - [Google Search Central blog](<https://developers.google.com/search/blog>): Get the latest information from our Google Search Central blog. You can find information about generative AI in Search, new Search Console features, and much more. - Google Search Central on [LinkedIn](<https://www.linkedin.com/showcase/googlesearchcentral/>) and [X (Twitter)](<https://twitter.com/googlesearchc>): Follow us for updates on Google Search and

resources to help you make a great site. | - [Google Search Central Help Forum](https://support.google.com/webmasters/community): Post questions about your site's SEO issues and find tips to create high quality sites from the product forum for website owners. There are many experienced contributors in the forum, including [Product Experts](https://productexperts.withgoogle.com/) and occasionally Googlers. - [Google Search Central YouTube Channel](https://www.youtube.com/c/GoogleSearchCentral): Watch hundreds of helpful videos created for website owners. |